Logistic Regression-Based Credit Scoring for Loan Default Reduction: A Quantitative Framework

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Abstract

Credit risk management remains one of the most critical challenges facing financial institutions globally. This paper presents a logistic regression-based credit scoring framework developed and validated on a portfolio of over 15,000 loan accounts. The model integrates borrower behavioural data, income metrics, and macroeconomic signals — including interest rate sensitivity analysis across a 13%–14.5% range — to produce a robust probability-of-default (PD) score. Implemented within a real lending environment, the framework achieved a 12% reduction in loan defaults and enabled data-driven segmentation of non-performing loan (NPL) cohorts exceeding a 7% threshold. Results demonstrate that structured quantitative scoring significantly outperforms judgment-based underwriting and provides a replicable foundation for credit risk automation in emerging market financial systems.

Keywords: *credit scoring · logistic regression · probability of default · NPL · credit risk · emerging markets · financial modelling*

1. Introduction

The extension of credit is a fundamental driver of economic growth. However, the misallocation of credit — particularly in emerging markets — leads to systemic financial instability, elevated non-performing loan (NPL) ratios, and reduced access to capital for creditworthy borrowers. Effective credit risk modelling is therefore not merely a technical exercise; it is a societal and economic imperative.

Traditional credit assessment in many markets still relies heavily on subjective judgement, limited financial history, and informal income verification. This approach is prone to inconsistency, bias, and error. The application of statistical learning methods — specifically logistic regression — offers a transparent, auditable, and scalable alternative.

This paper presents a complete logistic regression credit scoring framework developed for a portfolio of over 15,000 active loan accounts. The study addresses three core objectives:

- Construct a probability-of-default (PD) model using borrower-level financial and behavioural features.
- Quantify the sensitivity of loan uptake and default risk to interest rate movements.
- Segment borrowers by NPL risk and provide actionable credit limit recommendations.

2. Literature Review

2.1 Credit Scoring Models

Credit scoring has evolved significantly since the pioneering work of Altman (1968), who developed the Z-score model for bankruptcy prediction using linear discriminant analysis. Among subsequent approaches, logistic regression remains the gold standard in regulated financial environments due to its interpretability, regulatory acceptance under Basel II/III frameworks, and robust performance on structured tabular data (Siddiqi, 2006).

2.2 NPL Risk in Emerging Markets

The IMF (2023) reports that NPL ratios in Sub-Saharan Africa average 10.2%, nearly double the global average. Studies by Kipyego & Wandera (2013) highlight the particular challenge of credit risk assessment in Kenya, where mobile lending platforms have rapidly expanded credit access without commensurate risk controls.

2.3 Interest Rate Sensitivity

Stiglitz & Weiss (1981) demonstrated through their credit rationing model that raising interest rates does not unambiguously reduce credit risk — it may worsen the borrower pool through adverse selection, with direct implications for portfolio management during central bank rate adjustments.

3. Data and Methodology

3.1 Dataset

The analysis draws on a loan portfolio of 15,247 accounts observed over a 24-month period. The binary outcome variable Y = 1 if the account defaults (NPL > 90 days past due), 0 otherwise. The overall default rate was 9.4%, consistent with regional benchmarks.

Feature Category	Variables
Borrower Demographics	Age, employment type, region
Financial Profile	Monthly income, existing debt, savings balance
Loan Characteristics	Loan amount, tenure, interest rate, collateral
Behavioural Signals	Repayment history, missed payments, days-past-due
Macroeconomic	CBK benchmark rate, inflation index

3.2 Logistic Regression Model

The probability of default for account i is modelled as:

$$P(Y = 1 \mid x) = 1 / (1 + \exp(-(b0 + b'x)))$$

where x is the feature vector for borrower i and b are coefficients estimated by maximum likelihood.

3.3 Feature Engineering

Key engineered features include: Debt-to-Income Ratio (DTI = Total Obligations / Monthly Income); Repayment Consistency Index (RCI = proportion of on-time payments in prior 12 months); and Income-Loan Ratio (ILR = Loan Amount / Annual Income). Feature selection used Weight of Evidence (WoE), Information Value (IV > 0.10), and VIF analysis (threshold < 5).

3.4 Credit Scorecard

Coefficients were transformed into a point-based scorecard: Score = A - B × log(Odds), calibrated so that 600 corresponds to 50:1 good-to-bad odds.

Score Range	Risk Band	Recommended Action
720–850	Very Low	Approve; maximum credit limit
640–719	Low	Approve; standard limit
580–639	Medium	Approve with conditions
520–579	High	Reduce limit; enhanced monitoring
< 520	Very High	Decline or require collateral

4. Results

4.1 Model Performance

Metric	Training Set	Validation Set
AUC-ROC	0.821	0.808
Gini Coefficient	0.642	0.616
KS Statistic	0.412	0.389
Accuracy	88.3%	87.1%
Precision	0.74	0.71
Recall	0.69	0.67

An AUC of 0.808 on the holdout set indicates strong discriminatory power, well above the regulatory minimum of 0.70 under Basel IRB approaches.

4.2 Default Reduction Outcomes

Deployment of the scorecard resulted in a 12% reduction in the 12-month default rate (from 9.4% to 8.3%) relative to the prior judgement-based system. Additionally, 1,847 accounts in the NPL > 7% cohort were identified for targeted early intervention.

4.3 Interest Rate Sensitivity

Across the 13%–14.5% rate corridor, loan uptake elasticity was -1.34, indicating that a 1% increase in interest rates reduces new applications by approximately 1.34%. Default probability increased by an average of 0.6 percentage points per 50 basis point rate increase. Borrowers in the Medium risk band showed the highest sensitivity at 1.2 percentage points per 50 bps.

5. Discussion

The results confirm that a well-calibrated logistic regression scorecard can substantially improve credit portfolio quality in an emerging market context. The model achieves this while maintaining interpretability — each score point corresponds to an explicit combination of borrower characteristics, enabling credit officers to explain decisions to applicants and regulators alike.

The finding that Medium-risk borrowers are most sensitive to rate increases has direct implications for credit portfolio stress testing. Institutions should model CBK rate shock scenarios as standard components of their credit risk appetite framework.

6. Conclusion

This paper demonstrates that logistic regression-based credit scoring delivers meaningful improvements in loan portfolio quality when rigorously implemented with domain-appropriate feature engineering. The 12% default reduction achieved has direct implications for capital efficiency, provisioning costs, and responsible lending practices.

Future work includes: benchmarking against gradient boosting and neural networks; integrating M-Pesa transaction history as behavioural signals; and developing a dynamic credit limit adjustment mechanism triggered by real-time score updates.

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